8

Magnetic polarons and colossal magnetoresistance

Prior discussion addressed the effects of a carrier's energy depending on the positions of surrounding atoms. In a magnetic solid a carrier's energy also depends on exchange interactions that are governed by the relative alignments of a carrier's spin and unpaired spins of a host material's magnetic ions. Through these interactions a carrier influences the alignment of the host's unpaired spins. Meanwhile, the alignments of the host spins affect the carrier's state.

A *magnetic polaron* forms as a charge carrier added to an antiferromagnetic insulator (e.g. EuTe) self-traps inducing alignment of the associated spins into a ferromagnetic cluster. A magnetic polaron can be identified by its large and distinctive magnetic susceptibility. A high enough carrier density can even convert an antiferromagnet into a global ferromagnet. Here magnetic-polaron formation is addressed with a scaling analysis.

The scaling analysis is also applied to the free-energy arising from electron–phonon and exchange interactions of a donor electron in a ferromagnetic insulator (e.g. EuO). Exchange interactions between a large-radius donor electron and the localized spins of a ferromagnetic insulator oppose their thermally induced deviations from full alignment. At high enough temperature the cost in free energy of suppressing these spin deviations can become large enough to drive a donor to abruptly shrink to a bound small polaron with a concomitant loss of spin alignment. The temperature of this collapse rises in an applied magnetic field since it also suppresses spin deviations. Furthermore, direct electrical conduction between donors can be squelched by their collapse. In particular, with an appropriate density donors' thermally induced collapse can suppress their metallic impurity conduction to reveal the material's residual semiconducting behavior. The increase of the temperature of such a metal-to-semiconductor transition with an applied magnetic field is described as *colossal negative magnetoresistance* (CMR).

8.1 Magnetic polarons

A charge carrier usually self-traps by shifting the equilibrium positions of surrounding atoms. In principle a carrier could also self-trap by altering the ground-state alignment of the localized spins of a magnetic semiconductor (De Gennes, 1960). In particular, a carrier added to an antiferromagnetic insulator could realign associated spins to produce a ferromagnetic cluster. The proposed quasi-particle has been termed a "magnetic polaron" (Nagaev, 1967; Kasuya *et al*., 1970; Umehara and Kasuya, 1972) and a "ferron" (Nagaev, 1976).

Magnetic-polaron formation is indirectly driven by the intra-site exchange interaction between a carrier and the magnetic ion it occupies. The direct effect of this interaction is to foster parallel or anti-parallel alignment of the carrier's spin with that of the occupied magnetic ion. As a result, misalignment of the local moments of different magnetic ions impedes carriers' movement between them (Anderson and Hasegawa, 1955). In particular, when the net intra-site exchange energy is much larger than the inter-site transfer energy the carrier's spin will align with that of the occupied ion. Then electronic transfer between two magnetic ions of spin S is reduced by a factor which falls from unity to $1/(2S + 1)$ as their misalignment is increased. In the classical limit ($S \to \infty$) this reduction factor corresponds to $\cos(\theta/2)$, where θ is the misalignment angle. In the complementary limit, when the net intra-site exchange energy is much less than the intersite transfer energy, the carrier maintains its spin alignment as it moves between magnetic ions. As a result, a carrier aligned with the magnetic moments of a ferromagnetic cluster encounters an energy barrier upon attempting to exit the ferromagnetic cluster. All told, intra-site exchange interactions tend to confine a carrier to a ferromagnetic cluster embedded within an antiferromagnetic insulator.

Antiferromagnets form with equal and opposite alignments of the spins of adjacent (1) ferromagnetic planes, (2) ferromagnetic chains or (3) individual magnetic ions. These ferromagnetic structures have dimensionalities of $d_f = 2$, 1, or 0, respectively. With magnetic-polaron formation a carrier converts some antiferromagnetically aligned spins to ferromagnetic alignment. A saturated ferromagnetic core might be surrounded by a halo of gradually diminishing unsaturated ferromagnetic alignments (Mauger and Mills, 1985).

For simplicity nonetheless presume a sharp demarcation between the region of saturated carrier-induced ferromagnetic alignment and the host material's antiferromagnetism as illustrated in Fig. 8.1. The ground-state energy functional of such a magnetic polaron in an electronically isotropic antiferromagnetic insulator with d_f-dimensional ferromagnetic constituents is modeled as the sum of three contributions (Emin and Hillery, 1988):

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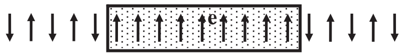

Fig. 8.1 A magnetic polaron (in the rectangle) forms in an antiferromagnetic insulator as a carrier of charge *e* induces ferromagnetic alignment of the spins surrounding it.

$$E_{\mathrm{mag},d_f}\left(L_a, L_f\right) = \frac{T_\mathrm{e}}{3}\left(\frac{3-d_f}{L_a^2} + \frac{d_f}{L_f^2}\right) - \frac{|I|S}{2} + J_{\mathrm{ex}}S^2\left(L_a^{3-d_f}L_f^{d_f}\right). \tag{8.1}$$

In Eq. (8.1) L_a and L_f are respectively the carrier's spatial extents (measured in units of the lattice constant) perpendicular to and parallel to an antiferromagnet's ferromagnetic planes, $d_f = 2$, or chains, $d_f = 1$. In addition, T_e is the carrier's half-bandwidth; I is the on-site exchange energy linking the spin-1/2 carrier with an occupied magnetic ion with spin magnitude S; and J_{ex} is the antiferromagnetic inter-ion exchange energy. The first contribution in Eq. (8.1) depicts the increase of a carrier's kinetic energy that results from containing it within the region of aligned spins. The second contribution is the net reduction of the energy of the carrier resulting from its intra-site exchange interactions with ferromagnetically aligned magnetic moments. This term is independent of the cluster's size because a carrier's interaction with each local moment is inversely proportional to the cluster size. The final contribution is the net increase of inter-ion exchange energy arising from flipping some of the antiferromagnet's spins to produce a ferromagnetic cluster.

The ground-state energy is found by minimizing the magnetic-polaron energy functional Eq. (8.1) with respect to L_a and L_f. At the minimum

$$L_a = L_f = L_{\mathrm{mag}} \equiv \sqrt[5]{2T_\mathrm{e}/3J_{\mathrm{ex}}S^2} \tag{8.2}$$

with

$$E_{\mathrm{mag}} = -\frac{|I|S}{2} + \frac{5}{3}\frac{T_\mathrm{e}}{L_{\mathrm{mag}}^2}. \tag{8.3}$$

The magnetic polaron's spatial extent L_{mag} is determined by the ratio of the carrier's confinement energy (characterized by T_e) to the inter-ion exchange energy $J_{\mathrm{ex}}S^2$.

The magnetic polaron will be energetically stable if E_{mag} is less than the net energy in the absence of carrier-induced realignments of the antiferromagnetic's local moments. In the limit $T_\mathrm{e} \gg |I|S/2$ the carrier maintains its spin as it moves between oppositely aligned sites. A carrier in the unaltered antiferromagnet is then

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delocalized with energy $E_{free} = 0$. The magnetic-polaron's stability condition becomes just $E_{mag} < 0$, where Eq. (8.3) and the condition $T_e \gg |I|S/2$ insures that the magnetic polaron be large: $L_{mag} > (10T_e/3\,|I|\,S)^{1/2} \gg 1$. Expressed alternatively, magnetic-polaron formation typically requires especially weak antiferromagnetic exchange:

$$T_e > \frac{|I|S}{2} > \frac{5}{3}\left[T_e^3\left(3J_{ex}S^2/2\right)^2\right]^{1/5}. \tag{8.4}$$

Applying a magnetic field tends to resize and reorient magnetic polarons' ferromagnetic clusters (Emin and Hillery, 1988). These effects yield huge magnetic susceptibilities for large ferromagnetic clusters. Thus a magnetic polaron should produce a very large magnetic susceptibility. Such an observation in doped EuTe is attributed to magnetic-polaron formation induced by wide *s*-band carriers' strong exchange with $S = 7/2$ *f*-band local moments within a very low-Néel-temperature antiferromagnet, $T_N \approx 9$ K (Vitins and Wachter, 1975). However, these observations may also indicate a ferromagnetic cluster formed by a carrier bound to a dopant, termed a bound magnetic polaron (BMP), rather than an intrinsic magnetic polaron (Mauger and Mills, 1985).

Electron–phonon interactions increasingly affect a magnetic polaron as its size is diminished (Umehara, 1981; Emin and Hillery, 1988). Indeed, like the situations described in Section 4.5, sufficient electronic localization of a magnetic-polaron's carrier can trigger its collapse into a small polaron. Similar effects come to the fore in the next section. Thermally produced spin deviations can destabilize a bound magnetic polaron in a ferromagnet with respect to its collapsing into a bound small polaron.

8.2 Donor-state collapse in ferromagnets: colossal magnetoresistance

Electron–phonon interactions can be pivotal in producing dramatic electronic transitions in magnetic semiconductors. In particular, as the local moments of a ferromagnetic semiconductor increasingly deviate from full ferromagnetic alignment a donor's electron ground state can abruptly collapse from being large to being severely localized. Such a sudden transition can occur because a short-range electron–phonon interaction only permits an extreme dichotomy in a carrier's localization. That is, a donor's electron can be so extended that the electron–phonon interaction has only a modest effect on its localization. Alternatively, the electron–phonon interaction can induce the donor's electron collapse to the smallest size compatible with a solid's atomicity and chemistry.

Consider a large-radius shallow donor added to a ferromagnetic insulator. Here exchange interactions between the magnetic insulator's local magnetic moments induce their ferromagnetic alignment. The exchange interactions of the donor electron with the ferromagnet's magnetic ions also foster their ferromagnetic alignment. The drive toward ferromagnetic alignment is therefore greatest in the vicinity of the donor electron. As a result, as depicted in Fig. 8.2, the donor electron tends to forestall thermally induced deviations of the local moments from full ferromagnetic alignment.

As discussed in Chapter 4, the short-range component of the electron–phonon interaction supports two qualitatively different ground states. In one instance the carrier extends over multiple sites. In the other situation the carrier collapses to the smallest size compatible with the solid's local chemistry. Direct extension of the scaling analysis of Section 4.3 yields the energy functional governing the ground state of a donor electron in a non-magnetic solid:

$$E_{\text{donor}}(L) = \frac{T_{\text{e}}}{L^2} - \frac{(V_{\text{L}} + V_{\text{donor}})}{L} - \frac{V_{\text{S}}}{L^3}. \tag{8.5}$$

Here the scaling constant produced by the donor electron's Coulomb attraction to its center is

$$V_{\text{donor}} \equiv C_{\text{donor}} \int d\boldsymbol{r} |\phi_n(\boldsymbol{r})|^2 \left(\frac{1}{r}\right) \tag{8.6}$$

and C_{donor} governs the strength of this attraction. The donor collapses when the minimum of the energy functional occurs at the smallest acceptable value of

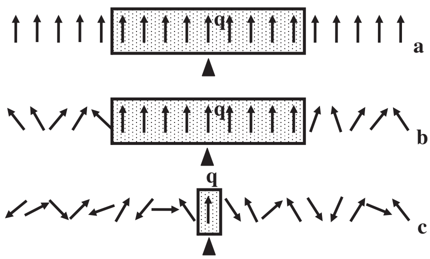

Fig. 8.2 The thermally induced collapse of a bound magnetic polaron in a ferromagnetic semiconductor is illustrated. (a) At very low temperatures the carrier (in the rectangle) bound to a defect (black triangle) does not perturb the alignment of the ferromagnet's spins. (b) At higher temperatures the carrier suppresses thermally induced deviations of surrounding spins from ferromagnetic alignment. (c) At high enough temperatures the carrier collapses thereby releasing surrounding spins to deviate from ferromagnetic alignment.

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L, $L = L_c$. The donor's ground state is extended when the minimum of the energy functional occurs at a larger value of L.

A magnetic solid's donor electron also has intra-site exchange interactions with the local magnetic moments it contacts. Through these interactions the electron of this bound magnetic polaron (BMP) limits magnetic ions' deviations from full ferromagnetic alignment. In particular, the donor electron suppresses magnetic ions' thermal deviations from ferromagnetic alignment. As depicted in Fig. 8.2 the suppression is greatest for large-radius states. Thus, as the ferromagnet's temperature is raised toward its Curie temperature the reduction of the free energy from magnetic ions' deviations from ferromagnetic alignment increasingly fosters reducing the donor electron's spatial extent.

The free energy has been calculated with the ferromagnet's spin deviations being described as spin waves (Emin *et al*. 1986; 1987), valid below about 75% of the Curie temperature (Low, 1963). That portion of the calculated free energy which depends upon the spatial extent of the donor electron is then given by

$$\mathrm{F}_{\mathrm{BMP}}(L,T) = \frac{T_{\mathrm{e}}}{L^2} - \frac{(V_{\mathrm{L}} + V_{\mathrm{d}})}{L} - \frac{V_{\mathrm{S}}}{L^3} - C\left(\frac{I}{\kappa T_{\mathrm{C}}}\right)^2\left(\frac{\kappa T}{L^2}\right), \tag{8.7}$$

where T_C is the ferromagnet's Curie temperature and C is a numerical constant. The temperature-dependent contribution to $F_{BMP}(L,T)$ drives the BMP to shrink with rising temperature.

If the short-range component of the electron–phonon interaction is strong enough it produces a dichotomy between large-radius and collapse minima. Then, as illustrated in Fig. 8.3, the free-energy minimum can abruptly shift from that of a large-radius donor to that of a collapsed donor as the temperature is raised (Emin *et al*., 1986; 1987). This thermally induced sudden collapse of a BMP is also depicted schematically in the lowest panel of Fig. 8.2. The difference between the entropy associated with spin deviations for large-radius and collapsed states underlies this thermally induced transition. The product of this entropy difference and the transition temperature is the latent heat accompanying the BMP's sudden collapse.

Applying a magnetic field suppresses spin deviations and thereby raises the temperature of the BMP collapse. In particular, the temperature of the collapse rises as the square root of the applied field (Hillery *et al*., 1988). As a result, even a modest field can significantly raise the temperature of the collapse.

The collapse of BMP donors can dramatically affect direct transport between them. In particular, insulating behavior always prevails if the concentration of BMP donors is sufficiently small. Furthermore, metallic transport always results for large enough concentrations of BMP donors. At intermediate concentrations the

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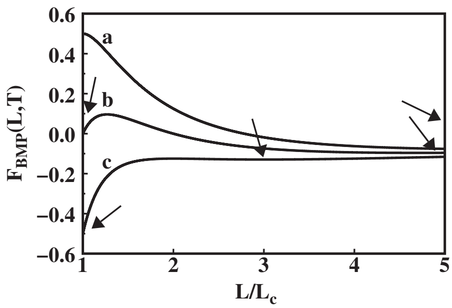

Fig. 8.3 The free-energy functional for a bound magnetic polaron in a ferromagnetic semiconductor, Eq. (8.7), is plotted against the ratio of the carrier's spatial extent L to that for a collapsed carrier L_c at three successively higher temperatures: curves a, b and c, respectively. Arrows point to the minima of these curves. The minimum at $L > L_c$ shrinks with rising temperature. Meanwhile the collapsed state's minimum $L = L_c$ emerges and is then stabilized.

thermally induced collapse of BMPs can shift their electrical transport from metallic to insulating. Applying a magnetic field raises the temperature of the collapse and the corresponding metal-to-insulator transition. At temperatures somewhat above the zero-field metal-to-insulator transition temperature application of the magnetic field thereby converts an insulator to metallic conduction. In other words, the rise of the metal-to-insulator transition temperature with magnetic field strength produces a huge negative magnetoresistance.

This discussion illustrates how electron–phonon interactions might contribute to abrupt transitions that occur in ferromagnetic semiconductors. In particular, this approach may be relevant to the dramatic metal-to-insulator transition observed in doped EuO.

Undoped EuO is a ferromagnetic semiconductor with a Curie temperature of 69 K (Methfessel and Mattis, 1968). It has the highest Curie temperature and smallest lattice constant of a series of structurally similar Eu-chalcogenide ferromagnetic crystals: EuO, EuS and EuSe. Ferromagnetism is thought to arise because direct ferromagnetic exchange between magnetic rare-earth ions dominates oxygen-mediated antiferromagnetic super-exchange. By contrast, antiferromagnetism dominates for EuTe which has the largest lattice constant.

N-type doping of EuO occurs through the introduction of oxygen vacancies (Oliver *et al.*, 1970; Torrence *et al.*, 1972; Godart *et al.*, 1980). Donors are also introduced by the substitution of Gd for Eu (Godart *et al.*, 1980). Samples that are too highly doped (> 1%) manifest metallic conduction without a metal-to-insulator transition (Godart *et al.*, 1980). Samples with intermediate doping levels show a dramatic transition from a low-temperature conducting state to a high-temperature insulating state at about 50 K, significantly below the Curie temperature (Oliver *et al*., 1970; Penny *et al*., 1972; Torrence *et al*., 1972; Oliver *et al*., 1972; Godart *et al*., 1980). While Hall-effect measurements imply that the mobility falls significantly upon entering the insulating state, a huge reduction of the carrier density accounts for most of the dramatic (13–15 order-of-magnitude) reduction of the electrical conductivity which occurs at the sharpest metal-to-insulator transition (Oliver *et al*., 1970; Petrich *et al*., 1971; Godart *et al*., 1980).

The thermally induced metal-to-insulator transitions observed in *n*-type EuO have been modeled as being thermally induced Mott-transitions in which carriers increasingly localize at donors (Torrence *et al*., 1972; Leroux-Hugon, 1972; Mott, 1974; Kübler and Vigren, 1975; Mauger, 1983). These transitions also have been interpreted as being due to the suppression of metallic impurity conduction which accompanies donor states' collapse (Emin, 1986; Emin *et al*., 1987; Hillery *et al*., 1988). As illustrated in Fig. 8.2, thermally induced carrier localization in a ferromagnet lowers the free energy by enabling local magnetic moments to deviate from ferromagnetic alignment. As described in Fig. 8.3, the short-range component of the electron–phonon interaction enhances the severity of carriers' localization and thereby sharpens the transition.